Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 02)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

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0 1 . 2 Explain how soil organisms provide services that humans depend on.

[3 marks]

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0 1 . 3 Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

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0 1 . 4 Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

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0 2 . 1 Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

0 2 . 2 Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

0 2 . 3 Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

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0 2 . 4 Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

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0 3 Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	**1900**	**1930**	**1960**	**1986**	**2020**
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

0 3 . 1 Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

Answer ..... %

0 3 . 2 Explain why whale populations were so easily over-exploited.

[3 marks]

Do not write outside the box

0 3 . 3 Describe how the International Whaling Commission regulates whaling.

[4 marks]

0 3 . 4 Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

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0 4 . 1 Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

Square-mesh escape panels

Minimum and maximum catchable sizes

No-take zones

Closed seasons

0 4 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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0 5 A new lowland nature reserve is being created on former farmland next to an existing ancient woodland.

0 5 . 1 Explain why the **area** and the **shape** of the reserve both matter in its design.

[3 marks]

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0 5 . 2 Explain the value of a biological corridor linking the new reserve to the ancient woodland.

[2 marks]

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0 5 . 3 Explain how each of the following abiotic factors could be managed to increase the biodiversity of the reserve.

[4 marks]

Water levels .....

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Soil nutrient levels .....

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Topography .....

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Light levels .....

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0 5 . 4 Much of the reserve will be kept as a **plagioclimax**. Explain what this means and how grazing is used to maintain one.

[3 marks]

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0 5 . 5 Explain why the removal or culling of some species may be necessary on the reserve.

[3 marks]

15

0 6 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 6 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 7 Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

0 7 . 1 Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

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0 7 . 2 Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

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0 7 . 3 Explain why a high biodiversity increases the ecological stability of a woodland.

[2 marks]

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0 8 . 1 State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

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0 8 . 2 Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

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0 8 . 3 Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

0 8 . 4 Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

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0 9 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 9 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 9 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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2 .....

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1 0 . 1 Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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1 0 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

1 1 . 2 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS